Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following hydroxide is acidic?

(A) $Al(OH)_3$

(B) $Ca(OH)_3$

(C) $Tl(OH)_3$

(D) $B(OH)_3$

Q.2

Boric acid is a weak monobasic acid and acts as Lewis acid

(A) By donating H^+

(B) By accepting OH^-

(C) By donating lone pair of electrons

(D) By accepting lone pair of electrons.

Q.3

Borax is converted into amorphous boron by the following steps:

$$\text{Borax} \xrightarrow{X} H_3BO_3 \xrightarrow{\Delta} B_2O_3 \xrightarrow[\Delta]{Y} B$$

X and Y are respectively:

(A) HCl, Mg

(B) HCl, C

(C) C, Al

(D) HCl, Al

Q.4

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Identify X and Y in the following reaction

$$BCl_3 + NH_4Cl \xrightarrow{140^oC/C_6H_5Cl} X \xrightarrow{NaBH_4} Y$$

(A) $X = NaBO_2, Y = B_2O_3$

(B) $X = Na_2B_4O_7, Y = H_3BO_3$

(C) $X = BN, Y = [NH_4]^+[BCl_4]^-$

(D) $X = B_3N_3H_3Cl_3, Y = B_3N_3H_6$

Q.5

Which is not the use of orthoboric acid?

(A) As an antiseptic and eye wash.

(B) In glass industry.

(C) In glazes for pottery.

(D) In borax - bead test.

Q.6

The factor responsible for weak acidic nature of B-F bonds in BF_3 is

(A) large electronegativity of fluorine

(B) three centred two electron bonds in BF_3

(C) $p\pi - d\pi$ back bonding

(D) $p\pi - p\pi$ back bonding

Q.7

Choose the correct sequence for the geometry of the given molecules Borazone, Borazole, $B_3O_6^{3-}$ ['P' stands for planar and 'N' stands for non-
planar]

(A) NP, NP, NP

(B) P, P, NP

(C) NP, P, NP

(D) NP, P, P

Q.8

In which of the following, a salt of the type KMO_2 is obtained?

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(A) $B_2H_6 + KOH(aq) \rightarrow$

(B) $Al + KOH(aq) \rightarrow$

(C) Both

(D) None

Q.9

In the following series of reactions:

i $A + 2HCl + 5H_2O \longrightarrow 2NaCl + X$

ii $X \xrightarrow[370\ K]{\Delta} HBO_2 \xrightarrow[\text{Red heat}]{\Delta} Z$

iii $4BF_3 + 3LiAlH_4 \longrightarrow Y + 3LiF + 3AlF_3$

The compounds A, X, Z and Y may be selected from following options

(A)

A	X	Z	Y
$Na_2B_4O_7$	H_3BO_3	B_2O_3	B_2H_6

(B)

A	X	Z	Y
H_3BO_3	$Na_2B_4O_7$	B_2H_6	B_2O_3

(C)

A	X	Z	Y
B_2H_6	B_2O_3	H_3BO_3	$Na_2B_4O_7$

(D)

A	X	Z	Y
B_2O_3	H_2BO_3	$Na_2B_4O_7$	B_2H_6

Q.10

$NaBH_4 + I_2 \rightarrow X + Y + Z$

$BF_3 + LiAlH_4 \xrightarrow{450K} X + P$

$BF_3 + LiAlH_4 \rightarrow X + Q + R$

X, Y, Z, P, Q and R in the reactions are

(A)

X	Y	Z	P	Q	R
$Na_4B_4O_7$	NaI	HI	HF	AlF_3	LiF

(B)

X	Y	Z	P	Q	R
B_2H_6	NaI	H_2	NaF	LiF	AlF_3

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(C)

X	Y	Z	P	Q	R
B_2H_6	BH_3	NaI	$B_3N_3H_6$	Al_2F_6	AlF_3

(D)

X	Y	Z	P	Q	R
BH_3	B_2H_6	H_2	$B_3N_3H_6$	LiF	AlF_3

Q.11

When an inorganic compound X having $3c-2e^-$ as well as $2c$ - $2e^-$ bonds react with NH_3 gas at a certain temperature, gives a compound Y, isostructural with benzene. Compound X with ammonia at a high temperature produces a substance Z, then

(A) X is BH_3, Y is $B_2N_2H_3$, Z is inorganic Benzene

(B) X is B_2H_6, Y is $B_3N_3H_6$, Z is Boron nitride

(C) X is borax, Y is B_2O_3, Z is inorganic Benzene

(D) Reactions insufficient to predict

Q.12

Boric acid is polymeric due to

(A) its acidic nature

(B) the presence of hydrogen bonds

(C) its monobasic nature

(D) its geometry

Q.13

In borax, the number of $-$ OH group attached to boron atoms is

(A) 2

(B) 3

(C) 4

(D) 6

Q.14

The treatment of Mg_3B_2 with acids produces

(A) B_2O_3

(B) MgB_2O_4

(C) B_4H_{10}

(D) MgB_4O_7

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Q.15

Which of the following is tetraboric acid?

(A) HBO_2

(B) H_3BO_3

(C) $H_2\ B_4O_7$

(D) B_2O_3

Q.16

Which of the following is the correct formula of borax?

(A) $Na_2\ [\ B_4O_5(OH)_4] \cdot 2H_2O$

(B) $Na_2\ [\ B_4O_5(OH)_4] \cdot 4H_2O$

(C) $Na_2\ [\ B_4O_5(OH)_4] \cdot 6H_2O$

(D) $Na_2\ [\ B_4O_5(OH)_4] \cdot 8H_2O$

Q.17

Which of the following statements is not correct?

(A) Boric acid is a very weak monobasic acid

(B) Boric acid contains planar BO_3 units which are bonded together through hydrogen bonds forming a layer structure

(C) Boric acid is used as a food preservative

(D) Boric acid is a tribasic acid

Q.18

Which of the following statements is correct?

(A) Borax involves two boron atoms in sp^2 hybridization and two boron atoms in sp hybridization

(B) Borax involves two boron atoms in sp^2 hybridization and two boron atoms in sp^3 hybridization

(C) Borax involves all the four atoms in sp^2 hybridization

(D) Borax involves all the four atoms in sp^3 hybridization

Q.19

Which of the following equations is not correctly formulated?

(A) $Na_2\ [\ B_4O_5(OH)_4] \cdot 8H_2O + 2HCl \rightarrow 2NaCl + 4H_3BO_3 + 5H_2O$

(B) $2BN + 6H_2O \rightarrow 2H_3BO_3 + 2NH_3$

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(C) $H_3BO_3 \xrightarrow[-H_2O]{375\ K} \underset{\text{metaboric acid}}{HBO_2}; 4HBO_2 \xrightarrow[-H_2O]{435\ K} \underset{\text{tetraboric acid}}{H_2B_4O_7}; H_2B_4O_7 \xrightarrow[-H_2O]{\text{red heat}} \underset{\text{Boric oxide}}{2B_2O_3}$

(D) H_3BO_3 is a weak monobasic acid as it liberates hydrogen ions as $H_3BO_3 \rightarrow H^+ + H_2BO_3^-$

Q.20

Which of the following statements is not correct?

(A) Boron BF_3 involves sp^2 hybridization

(B) BF_3 acts as a Lews acid

(C) BF_3 is a volatile liquid

(D) forms an adduct with NH_3

Answers & Solutions

Q.1 Answer:

$B(OH)_3$

Solution:

$B(OH)_3$ is acid because it can take OH^- ions.
H_3BO_3 or $B(OH)_3 + OH^- \rightarrow B(OH)_4^-$

Q.2 Answer:

By accepting OH^-

Solution:

$$B(OH)_3 + 2H_2O \rightleftharpoons [B(OH)_4]^- + H_3O^+$$

Q.3 Answer:

HCl, Mg

Solution:

$$\text{Borax} \xrightarrow{HCl} H_3BO_3$$

$$B_2O_3 \xrightarrow[\Delta]{Al} B(\text{crystalline})$$

$$B_2O_3 \xrightarrow[\Delta]{Mg} B(\text{Amorphous})$$

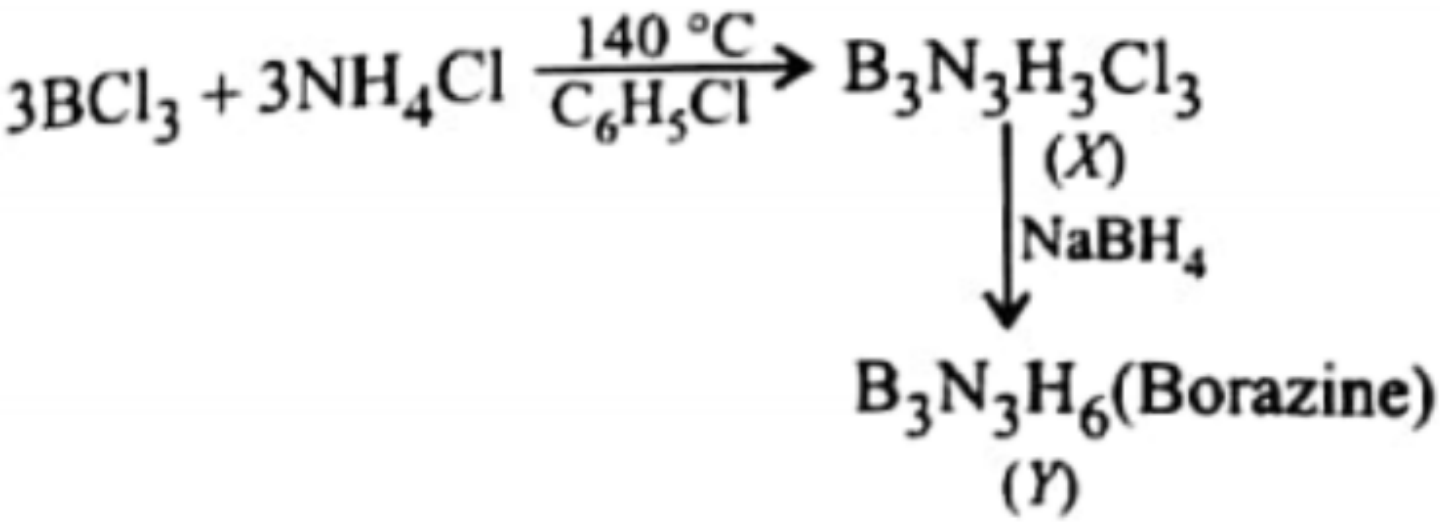
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Q.4 Answer:

$X = B_3N_3H_3Cl_3$, $Y = B_3N_3H_6$

Solution:

$X = B_3\ N_3H_3Cl_3, Y = B_3\ N_3H_6$



Q.5 Answer:

In borax - bead test.

Solution:

Borax on heating gives B_2O_3 and $NaBO_2$ which is glassy mass and used for boraxbead test.

Q.6 Answer:

$p\pi - p\pi$ back bonding

Solution:

It is $p\pi - p\pi$ back bonding involving B and F. The smaller atoms show more back bonding.

Q.7 Answer:

NP, P, P

Solution:

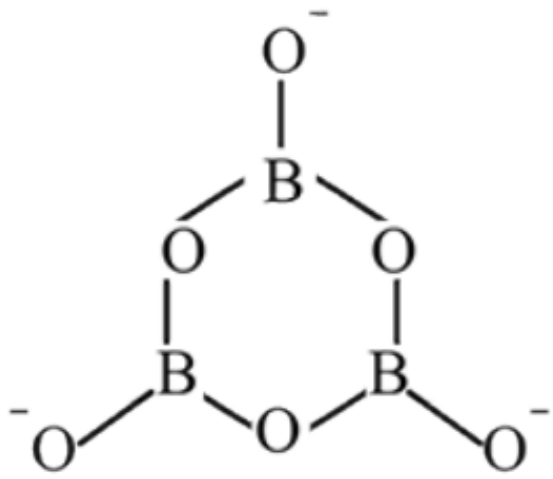
Borazone : A crystalline form of boron nitride which has diamond like structure. (Non-planar)

Borazole : Inorganic benzene : $B_3\ N_3H_6$ (planar)

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planar

$B_3O_6^{3-}$:



planar

Q.8 Answer:

Both

Solution:

$$B_2H_6 + 2KOH + 2H_2O \rightarrow \underset{\text{Potassium metaborate}}{2KBO_2} + 6H_2$$

$$2Al + 2KOH + 2H_2O \rightarrow \underset{\text{Potassium meta aluminate}}{2KAlO_2} + 3H_2$$

Q.9 Answer:

A	X	Z	Y
$Na_2B_4O_7$	H_3BO_3	B_2O_3	B_2H_6

Solution:

$$\underset{\text{Borax (A)}}{Na_2B_4O_7} + 2HCl + 5H_2O \longrightarrow 2NaCl + \underset{\text{Orthoboric aicd (X)}}{4H_3BO_3}$$

$$\underset{X}{H_3BO_3} \xrightarrow{\Delta,\ 370\ K} \underset{\text{Metaboric acid}}{HBO_2} + H_2O$$

$$4HBO_2 \xrightarrow{\Delta,\ \text{Red heat}} \underset{\text{Boron trioxide}}{2B_2O_3} + H_2O$$

z

$$4BF_3 + 3LiAlH_4 \rightarrow \underset{\text{(Diborane)}}{2B_2H_6} + 3LiF + 3AlF_3$$

Q.10 Answer:

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X	Y	Z	P	Q	R
B_2H_6	NaI	H_2	NaF	LiF	AlF_3

Solution:

$$2NaBH_4 + I_2 \rightarrow \underset{(X)}{B_2H_6} + 2NaI + \underset{(Z)}{H_2}$$

$$2BF_3 + 6NaH \xrightarrow{450K} \underset{X}{B_2H_6} + 6\underset{P}{NaF}$$

$$4BF_3 + 3LiAlH_4 \rightarrow \underset{(X)}{2B_2H_6} + \underset{(Q)}{3LiF} + \underset{(R)}{3AlF_3}$$

Q.11 Answer:

X is B_2H_6, Y is $B_3N_3H_6$, Z is Boron nitride

Solution:

This problem is based on preparation of inorganic benzene and structural charactristics of diborane. Student must be familiar with concept of bonding in diborane to solve this problem. Identify the compounds and complete the chemical reaction using information supplied in the question.

$$\underset{X}{B_2H_6} + 2NH_3 \text{excess} \xrightarrow[\Delta]{\text{Low temperature}} B_2H_6 \cdot 2NH_3 \xrightarrow[\Delta]{473K} \underset{\text{borozole Y}}{B_3N_3H_6}$$

Y has structure similar to benzene (inorganic benzene)

$$B_2H_6 + NH_3 \text{excess} \xrightarrow[\Delta]{\text{High temperature}} \underset{\text{boronnitride Z}}{BN_X}$$

Q.12 Answer:

the presence of hydrogen bonds

Solution:

In Boric acid each B atom is sp^2 hybridized and contains BO_3^{3-} units which are held together by hydrogen bonds.

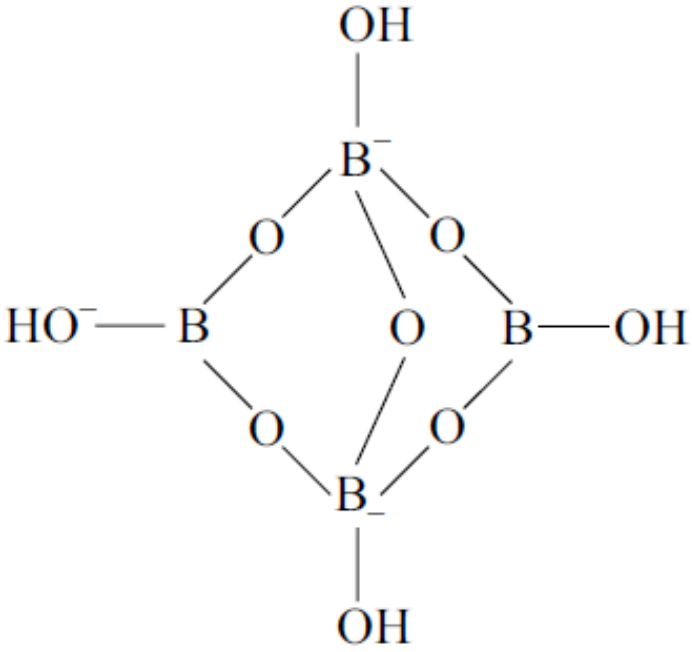
Q.13 Answer:

4

Solution:

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The structure of borax is



Q.14 Answer:

B_4H_{10}

Solution:

The treatment of borides with acids produces boranes.

Q.15 Answer:

$H_2\ B_4O_7$

Solution:

$H_2\ B_4O_7$

Q.16 Answer:

$Na_2\left[B_4O_5(OH)_4\right]\cdot 8H_2O$

Solution:

$Na_2\left[B_4O_5(OH)_4\right]\cdot 8H_2O$

Q.17 Answer:

Boric acid is a tribasic acid

Solution:

Boric acid is a tribasic acid

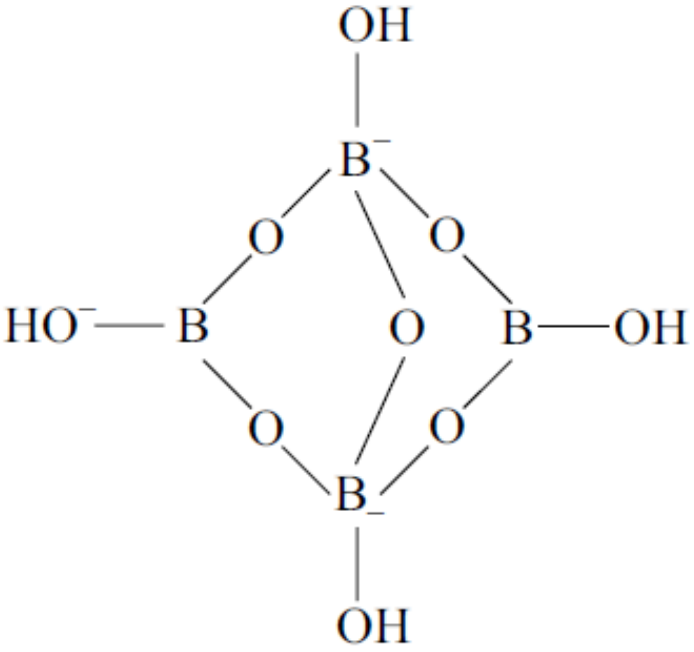
Q.18 Answer:

Borax involves two boron atoms in sp^2 hybridization and two boron atoms in sp^3 hybridization

Solution:

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The structure of borax is



Q.19 Answer:

H_3BO_3 is a weak monobasic acid as it liberates hydrogen ions as $H_3BO_3 \rightarrow H^+ + H_2BO_3^-$

Solution:

H_3BO_3 is a weak monobasic acid. It does not liberate H^+ but accepts OH^-, i.e. it is Lewis acid.

Q.20 Answer:

BF_3 is a volatile liquid

Solution:

BF_3 is a gaseous substance.

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